SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

KalaSetu

PROJECT OVERVIEW

- **Problem Statement ID – SIH26090**
- **Problem Statement Title- AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans**
- **Theme- Heritage and Culture**
- **PS Category- Software**
- **Team ID- GAT021**
- **Team Name (Registered on portal)- CodeLoom**



CodeLoom

KalaSetu

SMART INDIA HACKATHON 2026

Challenges and Problems

- **Lack of Digital Cataloguing Skills** – Traditional artisans may find it difficult to prepare professional digital product listings because of typing, literacy, and technical barriers.
- **Language and Regional Dialect Barriers** – Artisans' spoken descriptions may need conversion into structured product information and translated listings to reach a wider audience.
- **Poor Product Photography** – Product images may need background removal, lighting correction, and visual enhancement before being suitable for digital marketplaces.
- **Pricing Uncertainty** – Artisans may lack accessible guidance for determining appropriate pr prices, making feb and informed pricing difficult.
- **Dependence on Intermediaries and Temporary Trade Fairs** – Artisans may depend on middlemen and periodic exhibitions rather than maintaining continuous digital access to potential buyers.
- **Loss of Traditional Craft Knowledge** – The reasoning behind material selection and traditional techniques can remain undocumented, limiting the preservation of generational craft knowledge.

Proposed Solution

➢ **Voice-First AI Cataloguing**

Transforms an artisan's spoken description (including regional dialects) into a structured, professional product listing.

• Speech-to-text • AI-assisted listing generation • Hindi-English translation • Structured product fluids

➢ **AI Photo Studio and Dynamic Pricing**

Helps artisans prepare their products for digital presentation and provides pricing guidance.

• Background removal & cleanup • Lighting and colour correction • Dynamic Pricing Assistant (rule-based high- and category averages)

➢ **The "Why This Material" Wisdom Layer**

Captures the artisan's reasoning behind material and technique choices.

• Artisan explanations • Structured craft knowledge • Additional notes, stories and process videos (where supported)

Innovation and Uniqueness

VOICE-FIRST CATALOGUING

Artisans can describe a product verbally instead of typing. Spoken descriptions are converted into structured listings using AI.

MULTILINGUAL AI

Speech recognition and translation help convert regional language inputs into professional product descriptions in Hindi or English.

AI PHOTO STUDIO

Photo-enhancement tools assist with background cleanup, lighting correction and presentation of handmade products.

DYNAMIC PRICING ASSISTANT

Provides a suggested starting price using rule-based logic and category averages to reduce pricing uncertainty.

"WHY THIS MATERIAL" WISDOM LAYER

Captures the artisan's reasoning behind material selections and technique choices, documenting traditional craft knowledge alongside the product listing.

How It Addresses the Problems

- **Voice-First AI Cataloguing:** Reduces dependence on typing and manual data entry by transforming product product descriptions into structured listings.
- **Multilingual Auto-Cataloguing:** Helps bridge language barriers by generating or translating product descriptions into supported languages.
- **AI Photo Studio:** Assists with product photo preparation through background removal and visual corrections.
- **Dynamic Pricing Assistant:** Provides pricing guidance to help artisans make more informed initial pricing decisions.
- **Digital Marketplace Readiness:** Helps prepare product information for potential digital storefronts and marketplace channels, subject to implementation and integration requirements.
- **Wisdom Layer:** Documents the artisan's explanation of materials and techniques, helping preserve knowledge alongside commercial product information.

CodeLoom

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Methodology and Process

- 1. **Voice Capture and Speech Recognition:** Capture an artisan's spoken craft description in their regional dialect. Convert the speech into text using **AI4Bharat IndicASR / IndicWhisper.**
- 2. **AI Catalogue Generation:** Process the transcript using **Claude API** or **GPT-4o-mini** to generate a structured, professional product listing with an appropriate title, description, and product details.
- 3. **Multilingual Translation:** Use **Bhashini API** and **IndicTrans2** to translate craft descriptions and product listings into Hindi and English.
- 4. **Craft Wisdom Extraction:** Capture the artisan's explanation of why a particular material or technique was chosen. Use the AI processing stage to document this knowledge in the distinctive **"Why This Material" Wisdom Layer.**
- 5. **AI Photo Studio and Dynamic Pricing:** Use **rembg, OpenCV,** or **Remove.bg** for product-photo cleanup and background removal. Suggest fair product prices using rule-based methods, category averages, and potential **scikit-learn / XGBoost** models.
- 6. **Marketplace Integration and Deployment:** Prepare a complete product listing with a clean photograph, multilingual description, and suggested price for the KalaSetu storefront and intended integrations with **GeM and Amazon Karigar.**

Technology Stack

1. AI and Language Processing
AI4Bharat (IndicASR / IndicWhisper) · Bhashini · IndicTrans2 · Claude API · GPT-4o-mini

2. Image Processing
rembg · OpenCV · Remove.bg

3. Machine Learning & Pricing
learn · XGBoost

4. Mobile App Development
Flutter · React Native

5. Backend Development
Python · FastAPI · node

6. Databases and Storage
Firebase Firestore · MongoDB · Firebase Storage

7. Marketplace Integrations
GeM (Government e Marketplace) · amazon karigar

8. Development and Design Tools
GitHub · Figma

User Interface / System Architecture

ARTISAN INPUT & VOICE CAPTURE
- Artisan opens the KalaSetu mobile application.
- Records craft description in regional dialect.
- Selects language and dialect.
- Can review or edit the speech transcript.
- Optional typed description or craft-process video.

↓

SPEECH PROCESSING & BACKEND
- AI4Bharat IndicASR / IndicWhisper (speech-to-text).
- Bhashini API (language processing).
- Python FastAPI or Node.js (backend).
- Request handling and pipeline coordination.

↓

AI CATALOGUE & CRAFT WISDOM PLATFORM
- Claude API / GPT-4o-mini (structured listing).
- IndicTrans2 (Hindi & English translation).
- Product titles, descriptions and SEO-ready listings.
- "Why This Material" Wisdom Layer.
- Documentation of generational craft knowledge.

The Why Behind the Craft

Spoken Craft Story → Structured Listing + Material Knowledge

↓

AI PHOTO STUDIO & DYNAMIC PRICING

AI Photo Studio
- rembg / Remove.bg
- OpenCV (cleanup & lighting)

Dynamic Pricing Assistant
- Rule-based pricing & category avg.
- scikit-learn / XGBoost (optional)
- Suggested fair price

↓

PRODUCT LISTING & MARKETPLACE OUTPUT
- Complete product title and description.
- Hindi / English listing.
- Cleaned product photograph.
- Suggested fair price.
- KalaSetu in-app storefront.
- GeM & Amazon Karigar integrations.

VOICE → LISTING → PHOTO → FAIR PRICE → DIGITAL MARKETPLACE

Marketplace-Ready Artisan Product

Storage & Integrations
- Firebase Firestore or MongoDB
- Firebase Storage (images & assets)
- Bhashini API (language processing)
- GeM API (proposed integration)
- Amazon Karigar API (proposed integration)
- GitHub (source code mgmt)

CodeLoom

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Feasibility and Viability

Technical Feasibility

Voice-First AI Cataloguing
- AI4Bharat IndicASR / IndicWhisper for regional-language speech recognition.
- Claude API / GPT-4o-mini for structured product descriptions.
- Bhashini API and IndicTrans2 for multilingual processing.
- Python FastAPI or Node.js for backend development.

AI Product Enhancement
- rembg, OpenCV, or Remove.bg for product-photo background removal and cleanup.
- Rule-based pricing and category averages as an initial pricing baseline.
- Potential scikit-learn / XGBoost models for future pricing improvements.

Technical Risks
- Speech recognition quality may vary across regional dialects and recording conditions.
- AI-generated product descriptions require artisan review and correction.
- Internet connectivity, API availability, and external service reliability must be considered.

Financial Feasibility

Cost-Conscious Development
- Open-source tools can reduce initial software licensing costs.
- External AI APIs and cloud services may introduce usage-based expenses.
- Flutter or React Native can support the mobile application.

Scalable Infrastructure
- Firebase Firestore or MongoDB for application data.
- Firebase Storage for product photographs and associated assets.
- Start with a small pilot and expand infrastructure as usage increases.

Funding and Sustainability
- Explore relevant government grants and artisan-development programs.
- Investigate suitable institutional partnerships and CSR opportunities.
- Monitor API consumption, hosting costs, and ongoing maintenance expenses.

Operational Feasibility

Simple Artisan Workflow
- Voice-first interaction reduces dependence on typing.
- Guided steps help artisans create digital product listings.
- Users can review and correct AI-generated information.

End-to-End Product Processing
- Record the artisan's spoken craft description.
- Generate and translate the product listing.
- Clean the product photograph.
- Suggest a product price and prepare the listing for review.

Deployment and Maintenance
- Begin with a limited pilot.
- Provide onboarding and user support.
- Monitor technical errors, API reliability, and user feedback.
- Improve the application incrementally.

Social Viability

Inclusive Digital Access
- Voice-based interaction can reduce barriers associated with typing.
- Multilingual product descriptions can improve accessibility.
- A simple mobile workflow can help users with limited digital experience.

Preservation of Craft Knowledge
- The "Why This Material" Wisdom Layer documents traditional material and technique choices.
- Preserve craft context alongside product descriptions.
- Allow artisans to review and verify the authenticity of their knowledge.

Artisan Empowerment
- AI Photo Studio helps prepare cleaner product photographs.
- Dynamic Pricing Assistant provides a starting point for pricing decisions.
- Digital cataloguing prepares artisans for wider marketplace visibility.

Strategies for Overcoming Challenges

TECHNICAL CHALLENGES
- **Regional Language Accuracy:** Test speech recognition with representative artisan recordings, allow transcript corrections, and offer typed-input alternatives.
- **Low Connectivity:** Optimize image sizes, reduce unnecessary network requests, and investigate local draft saving for later synchronization.
- **AI and Integration Reliability:** Validate generated product information, handle API failures, and test integrations before deployment.

FINANCIAL CHALLENGES
- **API and Cloud Costs:** Start with a limited pilot, monitor usage, and establish spending limits for external AI services.
- **Affordable Infrastructure:** Prefer cost-effective cloud configurations and suitable open-source tools where practical.
- **Sustainable Expansion:** Explore appropriate grants, institutional partnerships, and CSR opportunities before planning large-scale rollout.

OPERATIONAL CHALLENGES
- **Artisan Onboarding:** Provide practical demonstrations, simple instructions, and regional-language guidance.
- **Trust and Accuracy:** Allow artisans to review descriptions, material information, photographs, and suggested prices before publishing.
- **Adoption and Maintenance:** Gather user feedback, provide support, and improve the application incrementally.

CodeLoom

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

BENEFITS OF KALASETU

FOR ARTISANS AND WEAVERS

- Voice-first cataloguing reduces dependence on typing and technical skills.
- AI-assisted product descriptions help artisans prepare professional listings.
- Photo enhancement and pricing guidance support product preparation.
- Digital cataloguing can help artisans reach audiences beyond temporary trade fairs and middlemen.

FOR CUSTOMERS AND BUYERS

- Structured product information helps buyers understand craft products.
- Multilingual descriptions improve access to craft stories.
- Documented materials, techniques, and regional context can build trust.
- More informative listings help buyers appreciate traditional craftsmanship.

FOR CULTURE AND HERITAGE

- The "Why This Material" Wisdom Layer captures artisans' explanations of material and technique choices.
- Craft stories preserve knowledge alongside commercial product information.
- Artisan notes, photographs, and process videos can document traditional practices.
- Regional identity and craft context remain connected to the product.

FOR THE DIGITAL ARTISAN ECONOMY

- Digital cataloguing supports year-round product presentation.
- AI-assisted descriptions can help artisans prepare marketplace-ready information.
- Pricing guidance provides a starting point for pricing decisions.
- Potential future marketplace integrations may expand digital visibility, subject to access and implementation.

Potential Impacts on the Target Audience

- **For Artisans:** Makes digital product cataloguing more accessible through voice-first interaction and guided listing preparation.
- **For the Economy:** Creates opportunities for direct digital product presentation and potential access to wider markets without relying exclusively on intermediaries.
- **For Culture:** Documents traditional materials, techniques, and artisan knowledge through the "Why This Material" Wisdom Layer.
- **For Society:** Supports more inclusive digital participation by reducing typing and language barriers for artisans with limited digital experience.
- **For Buyers:** Provides structured product information, multilingual descriptions, and richer craft stories that help communicate authenticity and context.
- **For Sustainability:** Encourages the continued visibility of traditional crafts and supports digital preservation of knowledge that may otherwise remain undocumented.

Business Model

FUNDING SOURCES FOR SUSTAINABILITY

GOVERNMENT PROGRAMS

Explore suitable government entrepreneurship and artisan-development initiatives.

Investigate relevant tourism, handicraft, and digital-inclusion schemes.

GRANTS AND INSTITUTIONAL SUPPORT

Explore eligible innovation grants and institutional partnerships.

Consider support from incubators and organizations working with artisans.

CSR AND PRIVATE PARTNERSHIPS

Explore potential corporate social responsibility support.

Investigate partnerships with technology providers and organizations supporting craft communities.

SUPPORTING SOURCES

PUBLIC-SECTOR SUPPORT
Potential collaboration with relevant government departments, subject to eligibility and approval.

CULTURAL AND CRAFT ORGANIZATIONS
Potential collaboration with artisan collectives, handicraft organizations, and cultural institutions.

TECHNOLOGY PARTNERSHIPS
Potential support through suitable cloud, AI, mobile technology, and digital-commerce partnerships.

TARGET CUSTOMERS

- Marginalized artisans and traditional craft producers.
- Handloom weavers and independent craft entrepreneurs.
- Artisan collectives and craft cooperatives.
- Organizations supporting handicraft preservation and digital inclusion.
- Buyers seeking authentic, well-documented craft products.

FINANCIAL MODEL

- Potential government schemes and eligible innovation grants.
- Possible CSR support and institutional partnerships.
- Potential partnerships with technology providers and craft organizations.
- A phased rollout to manage infrastructure and operating costs.

Funding and long-term operating arrangements require validation.

VALUE-ADDED SERVICES

- **Voice-First AI Cataloguing:** Converts spoken craft descriptions into structured product information.
- **Multilingual Listings:** Helps communiacte craft information in Hindi, English, and supported languages.
- **AI Photo Studio:** Assists with product-photo cleanup and presentation.
- **Dynamic Pricing Assistant:** Provides a starting price using rule-based logic and category averages, with potential future predictive models.
- **Wisdom Layer:** Preserves the artisan's explanations of materials, techniques, and craft traditions.

SCALABILITY

PILOT → IMPROVE → EXPAND

- Start with a small pilot involving a limited group of artisans.
- Refine the voice-first workflow using artisan feedback.
- Expand language support as speech-recognition quality permits.
- Add storage and processing capacity as demand grows.
- Investigate GeM and Amazon Karigar connections subject t to API access, eligibility, and integration requirements.

CodeLoom

KalaSetu

RESEARCH AND REFERENCES

Sample UI Development by the Team



KalaSetu Homepage



Kannada Translation



Hindi Translation

Reference App & Web

Indianhandmade
Government-supported artisan marketplace
indianhandmade.com

GoCoop / GoSwadeshi
Digital marketplace for handlooms and crafts
gem.gov.in

Okhai
Handcrafted products and artisan empowerment
okhai.org

External reference platforms for research and inspiration; not KalaSetu integrations.

Reference Articles & Sources

- **Digital Craft Marketplaces**
  Indianhandmade: A Digital Platform for India's Craft Heritage (verify metadata Hot assumed)
  https://www.pib.gov.in/PressReleasePage.aspx?PRID=2281569&lang=1&reg=48
- **Indian Language Technology**
  Towards Leaving No Indic Language Behind: Building Monolingual Corpora, Benchmark and Models for Indic Languages (ACL, 2023)
  https://aclanthology.org/2023.acl-long.693/
- **Digital Procurement**
  Government e-Marketplace (GeM), Government of India
  https://gem.gov.in/
- **Indian Language AI**
  AI4Bharat, IIT Madras
  https://ai4bharat.iitm.ac.in/
- **Translation Research**
  IndicTrans2, AI4Bharat
  https://github.com/AI4Bharat/IndicTrans2
- **Artisan Commerce**
  Amazon Karigar, Amazon India
  https://sell.amazon.in/grow-your-business/amazon-karigar

External references only; no partnership, endorsement, or integration implied.